

**THE IMPACT OF IMMERSIVE TECHNOLOGIES ON STEM
EDUCATION IN SRI LANKA'S**

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Final Dissertation

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DECLARATION

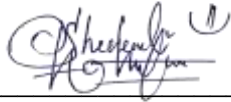
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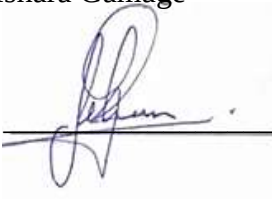
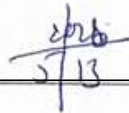
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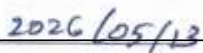
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ABSTRACT

This study investigates the effectiveness of Augmented Reality (AR)-based mobile learning in enhancing Grade 8 students' conceptual understanding of astronomy within the Sri Lankan secondary education context. Traditional instructional approaches rely on static textbook illustrations and verbal explanations, which often fail to convey the dynamic and three-dimensional nature of the Solar System, resulting in limited student engagement and spatial comprehension.

To address this limitation, a marker-based AR mobile application was developed to enable interactive visualization of planetary systems through real-time 3D models. The system is designed to function offline and is optimized for low-end Android devices, ensuring accessibility in resource-constrained educational environments.

A quasi-experimental approach was employed using a paired sample design, where the same group of students was evaluated under both traditional and AR-assisted learning conditions. Pre-test and post-test assessments were used to measure improvements in conceptual understanding and academic performance.

The results indicate a statistically significant improvement in student performance following the use of the AR application ($p < 0.05$), with notable gains in spatial understanding and learner engagement. These findings demonstrate the potential of AR as an effective tool for enhancing science education and contribute empirical evidence within the context of secondary education in Sri Lanka.

Keywords: Augmented Reality, Astronomy Education, Solar System, STEM Learning, Mobile Learning, Interactive Learning, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AR	Augmented Reality
STEM	Science, Technology, Engineering, and Mathematics
UI	User Interface
UX	User Experience
APK	Android Package Kit
FPS	Frames Per Second
RAM	Random Access Memory
SUS	System Usability Scale
DDE	Design–Development–Evaluation
MCQ	Multiple Choice Question
SD	Standard Deviation
t-test	Student’s t-Test
3D	Three-Dimensional
HCI	Human–Computer Interaction
VR	Virtual Reality
GPS	Global Positioning System
CPU	Central Processing Unit
MB	Megabyte

Android OS	Android Operating System
Vuforia SDK	Vuforia Software Development Kit
APK File	Android Application Package File
IEEE	Institute of Electrical and Electronics Engineers
SDLC	Software Development Life Cycle
CAD	Computer-Aided Design
AI	Artificial Intelligence
XML	Extensible Markup Language

CHAPTER 1: INTRODUCTION

1.1 Background and Literature Survey

1.1.1 The Sri Lankan Astronomy Education Context

Astronomy-related concepts in the Sri Lankan Grade 8 curriculum are delivered primarily through the geography syllabus prescribed by the Ministry of Education. Core topics include the structure of the Solar System, planetary characteristics, and spatial relationships between celestial bodies. The official textbook published by the Educational Publications Department serves as the principal learning resource for most students, particularly in government schools.

Despite the conceptual importance of astronomy, instruction remains largely dependent on static, two-dimensional diagrams and teacher-centered explanations. This approach requires students to mentally visualize complex three-dimensional relationships, which has been shown to be challenging for learners with limited spatial reasoning skills [1]. The issue is further intensified in under-resourced schools where access to supplementary visual aids or digital learning tools is minimal.

Additionally, the examination-oriented nature of the education system emphasizes memorization over conceptual understanding. As a result, students often struggle to grasp dynamic astronomical relationships such as planetary motion, relative scale, and positional alignment. These limitations highlight the need for more interactive and visualization-driven learning approaches.

1.1.2 Augmented Reality in Science and Astronomy Education

Augmented Reality has gained significant attention as an innovative educational technology capable of enhancing visualization and interactivity in learning environments. AR enables the overlay of virtual three-dimensional objects onto real-world contexts, thereby facilitating experiential and immersive learning [2].

A growing body of research indicates that AR-based learning environments improve student engagement, motivation, and conceptual understanding, particularly in science education [3]. By allowing learners to interact with virtual models in real time, AR supports active learning and strengthens the connection between abstract concepts and their visual representations.

In the context of astronomy education, AR has proven especially effective in enhancing spatial understanding, as it allows students to explore celestial structures in three

dimensions rather than relying on static diagrams [4]. However, many existing AR applications are developed for high-end devices and require continuous internet access, limiting their applicability in low-resource educational settings.

1.1.3 AR Applications for Solar System and Spatial Learning

Several studies have explored the use of AR applications to support the teaching of Solar System concepts. These applications enable students to visualize planetary structures, compare relative sizes, and understand spatial positioning through interactive 3D models.

Research findings suggest that students exposed to AR-based learning demonstrate significantly improved spatial reasoning and knowledge retention compared to those using traditional instructional methods [5]. Interactive features such as rotation, scaling, and real-time manipulation have been identified as key factors contributing to improved learning outcomes.

Furthermore, the integration of gamification elements within AR environments has been shown to enhance student engagement and motivation. Activities involving sequencing, comparison, and interactive problem-solving promote deeper cognitive processing and active participation [6]. These findings align with constructivist learning theories, which emphasize learning through interaction and exploration.

1.1.4 Challenges in AR Implementation in Resource-Constrained Environments

Despite the promising benefits of AR in education, several challenges limit its widespread adoption, particularly in developing countries. One major limitation is the dependency on high-performance devices, which are not readily available in many school environments [7].

Technical issues such as poor marker tracking, low frame rates, and application instability can negatively affect user experience and reduce educational effectiveness. Additionally, many AR solutions are not aligned with local curricula, making integration into classroom teaching practices difficult [8].

Another critical challenge is the lack of region-specific research. Most AR-based educational studies have been conducted in developed countries, raising concerns about the applicability of these findings to contexts such as Sri Lanka, where technological infrastructure and resource availability differ significantly [9].

These limitations highlight the need for AR applications that are specifically designed for low-resource environments, with features such as offline functionality, compatibility with low-end devices, and alignment with national curricula. Addressing these factors is essential for ensuring both usability and educational impact.

1.2 Research Gap

Despite the growing adoption of digital learning technologies and the proven effectiveness of Augmented Reality in enhancing visualization and engagement in education [1], [2], several limitations remain in its application within secondary-level astronomy learning. Many existing AR solutions are developed in controlled environments with advanced technological resources, limiting their applicability in real classroom settings, particularly in developing countries.

Furthermore, current implementations often lack alignment with national curricula and function primarily as supplementary tools rather than integrated learning solutions. In addition, limited attention has been given to optimizing AR applications for low-resource environments and evaluating their effectiveness using structured, quantitative methods.

Therefore, the following research gaps are identified in relation to AR-based astronomy learning systems.

Gap 1: Lack of curriculum-aligned AR applications for astronomy education

Most existing AR-based educational applications are designed as generic learning tools and are not directly aligned with specific national curricula [3]. In the context of Sri Lanka, where the Grade 8 astronomy syllabus is delivered through a structured textbook, the absence of curriculum-integrated AR solutions limits classroom adoption. Students are often required to switch between learning materials and digital applications, reducing instructional continuity. A gap exists in developing AR systems that are tightly integrated with textbook content, enabling seamless learning experiences through marker-based interactions.

Gap 2: Limited focus on Solar System-specific spatial learning

While AR has been widely applied in science education, a significant portion of research focuses on general STEM concepts or geometry-related visualization [2], [4]. Comparatively fewer studies specifically address astronomy topics such as planetary structure, relative scale, and spatial positioning within the Solar System. These concepts inherently require three-dimensional understanding, which traditional

teaching methods fail to adequately support. A gap exists in designing AR applications specifically targeted at improving spatial comprehension of Solar System concepts.

Gap 3: Lack of low-resource optimized AR solutions

Many AR applications are developed for high-end devices and require stable internet connectivity for full functionality [5]. In developing country contexts, including Sri Lanka, students often rely on low-end Android devices with limited processing power and intermittent connectivity. Applications that are not optimized for such conditions result in poor performance, reduced usability, and limited adoption. A gap exists in developing AR solutions that are lightweight, offline-capable, and compatible with low-end devices to ensure accessibility in resource-constrained environments.

Gap 4: Limited empirical evaluation using controlled comparison methods

Although several studies report positive outcomes of AR-based learning, many rely on qualitative feedback or observational data rather than rigorous quantitative evaluation [6]. There is a lack of studies employing structured experimental or quasi-experimental designs, such as pre-test/post-test comparisons, to measure actual learning improvement. A gap exists in conducting statistically validated evaluations that compare traditional learning methods with AR-assisted learning to provide empirical evidence of effectiveness.

Gap 5: Insufficient integration of interactive and gamified learning elements

Existing AR applications often focus primarily on visualization, with limited incorporation of interactive or gamified components that actively engage learners [7]. Research suggests that gamification can significantly enhance motivation, participation, and knowledge retention when combined with interactive technologies. However, few AR systems integrate structured learning activities such as comparison tasks, sequencing exercises, or interactive quizzes. A gap exists in combining AR visualization with gamified learning strategies to enhance both engagement and learning outcomes.

Gap ID	Limitation in Existing Approaches	Consequence	Research Need Addressed
G1	AR applications lack alignment with national curriculum	Reduced classroom integration and learning continuity	Curriculum-integrated AR system linked with textbook content
G2	Limited focus on Solar System-specific learning	Poor spatial understanding of astronomical concepts	AR application targeting Solar System visualization
G3	High dependency on advanced devices and internet	Limited accessibility in low-resource environments	Offline, low-end device optimized AR solution
G4	Lack of rigorous quantitative evaluation	Weak empirical validation of effectiveness	Statistical evaluation using pre-test/post-test analysis
G5	Minimal use of gamification in AR learning	Lower engagement and interaction levels	Integration of gamified learning activities within AR

Table 2: Summary of Research Gaps

1.3 Research Problem

Astronomy education at the Grade 8 level in Sri Lanka is primarily delivered through textbook-based instruction, where concepts such as planetary structure, spatial relationships, and relative motion are represented using static two-dimensional diagrams. While these materials provide foundational knowledge, they are often insufficient for conveying the dynamic and three-dimensional nature of the Solar System, leading to difficulties in student comprehension and reduced engagement [1]. Unlike subjects that can be easily demonstrated through physical experiments, astronomy relies heavily on visualization, making effective concept delivery particularly challenging in traditional classroom environments.

In practice, students are required to mentally interpret spatial relationships between celestial bodies, such as relative size, distance, and positioning. This cognitive demand can be difficult to meet, especially for learners with lower spatial reasoning abilities. Furthermore, the lack of interactive teaching tools and visual aids in many schools, particularly in resource-constrained environments, limits opportunities for experiential

learning. As a result, students often rely on memorization rather than conceptual understanding, which negatively impacts knowledge retention and application [2].

Although Augmented Reality has been widely recognized as an effective tool for enhancing visualization and engagement in science education, its application in classroom settings remains limited. Many existing AR-based learning systems are developed as standalone or supplementary tools that are not directly integrated with textbook content or aligned with specific curricular requirements [3]. This disconnect reduces their practicality for classroom use and limits adoption by educators.

Another significant challenge lies in the technical limitations of existing AR applications. A considerable number of solutions require high-performance devices and continuous internet connectivity, which are not consistently available in Sri Lankan school environments [4]. Applications that are not optimized for low-end devices often result in poor performance, including lag, tracking instability, and reduced usability, ultimately affecting the learning experience.

Furthermore, while prior studies highlight improvements in engagement and visualization through AR, many lack rigorous evaluation using structured quantitative methods such as controlled comparisons between traditional and technology-enhanced learning approaches [5]. This limits the availability of empirical evidence supporting the effectiveness of AR in improving actual learning outcomes within the local educational context.

To address these challenges, the proposed system focuses on the design and development of a marker-based AR mobile application tailored for Grade 8 astronomy education. The system is expected to:

- Enable interactive visualization of Solar System concepts through real-time 3D models
- Integrate directly with textbook content using marker-based interactions
- Operate offline and perform efficiently on low-end Android devices
- Incorporate interactive and gamified learning activities to enhance engagement
- Provide measurable improvements in conceptual understanding through structured evaluation

Based on the identified challenges, the central research problem is formulated as follows:

“How can a low-cost, offline, curriculum-aligned AR mobile application be designed and developed for Grade 8 astronomy education to improve students’ conceptual understanding, engagement, and spatial learning within resource-constrained classroom environments?”

1.4 Research Objective

This research focuses on enhancing the teaching and learning of astronomy concepts within the Sri Lankan Grade 8 curriculum through the development of a marker-based Augmented Reality mobile application. In traditional classroom settings, astronomy topics are delivered using static textbook diagrams, which often fail to effectively convey three-dimensional spatial relationships and dynamic planetary structures [1], [2]. As a result, students experience difficulties in conceptual understanding and engagement.

Although AR has demonstrated strong potential in improving visualization and interactivity in science education, existing solutions are often not aligned with national curricula, lack optimization for low-resource environments, and are rarely evaluated using structured quantitative methods [3], [4]. Therefore, the objective of this study is not only to develop an interactive AR-based learning system, but also to ensure that it is curriculum-aligned, accessible on low-end devices, and empirically validated within a real classroom context.

Main Objective

To design and develop a low-cost, offline-capable, curriculum-aligned AR mobile application for Grade 8 astronomy education and to evaluate its effectiveness in improving students’ conceptual understanding, spatial reasoning, and engagement within Sri Lankan secondary school classrooms.

Specific Objectives

RO1: Curriculum alignment of astronomy learning content

To map the Sri Lankan Grade 8 geography (astronomy) syllabus and textbook content to AR-based learning modules, ensuring alignment with topics such as the Solar System, planetary characteristics, and spatial relationships.

RO2: Development of a marker-based AR mobile application

To design and implement a mobile AR application using marker-based tracking to render interactive 3D models of astronomical objects, enabling real-time visualization of Solar System concepts.

RO3: Implementation of interactive 3D visualization features

To develop interactive functionalities such as rotation, scaling, and zooming, allowing students to explore and manipulate virtual planetary models to enhance spatial understanding.

RO4: Integration of gamified learning activities

To incorporate gamified components, including planet size comparison, planetary order sequencing, and fact-matching tasks, in order to improve student engagement and knowledge retention.

RO5: Optimization for low-resource environments

To ensure that the application operates efficiently on low-end Android devices and functions without continuous internet connectivity, making it suitable for resource-constrained school environments.

RO6: Empirical evaluation of learning effectiveness

To conduct a structured evaluation using a paired sample approach (pre-test and post-test) to measure improvements in students' conceptual understanding and academic performance when using the AR application compared to traditional learning methods.

RO7: Usability and engagement assessment

To assess user experience, engagement levels, and system usability through student interaction data and feedback collected during the evaluation phase.

Research Objective	Addresses G1	Addresses G2	Addresses G3	Addresses G4	Addresses G5
RO1: Curriculum alignment	✓	✓			
RO2: AR application development	✓	✓			
RO3: Interactive visualization		✓			✓
RO4: Gamification integration					✓
RO5: Low-resource optimization			✓		
RO6: Empirical evaluation				✓	
RO7: Usability assessment			✓	✓	✓

Table 3: Objectives vs Research Gaps

CHAPTER 2: METHODOLOGY

2.1 Methodology Overview

This chapter presents the methodology adopted to design, develop, and evaluate an Augmented Reality-based mobile learning system for Grade 8 astronomy education. The methodology is structured around three key aspects: (i) design and development of a marker-based AR application, (ii) integration of interactive and gamified learning features, and (iii) empirical evaluation of the system using a quantitative research approach.

The overall approach follows a Design–Development–Evaluation (DDE) framework, which is widely used in educational technology research to ensure both technical functionality and pedagogical effectiveness [3], [4], [16]. The methodology begins with system design and content mapping, followed by application development and

feature implementation, and concludes with structured evaluation using statistical analysis.



Figure 1: Overall Methodology Pipeline(DDE Framework)

2.1.1 System Context and Architecture Design

The proposed system adopts a layered and modular architecture, as the AR-based learning process involves multiple interconnected components, including user interaction, AR rendering, application logic, and data management [2], [3], [17]. This architectural approach ensures maintainability, scalability, and efficient performance, particularly on low-end Android devices commonly available in resource-constrained educational environments [7], [8].

The system architecture is composed of four primary layers. The presentation layer manages the mobile user interface and AR visualization components, enabling students to interact with the application in an intuitive manner [3], [4]. The application layer handles lesson logic, gamified learning activities, and interaction control mechanisms required for user engagement [15], [18]. The AR engine layer is responsible for real-time rendering and marker tracking through the integration of Unity and Vuforia technologies, allowing virtual 3D models to be accurately overlaid onto textbook markers [13], [14]. The data layer manages the local storage of quiz results, interaction records, and user performance data to support offline functionality and learning evaluation [8].

This modular structure enables independent development, testing, and maintenance of each component while ensuring seamless integration across the overall system architecture [17].

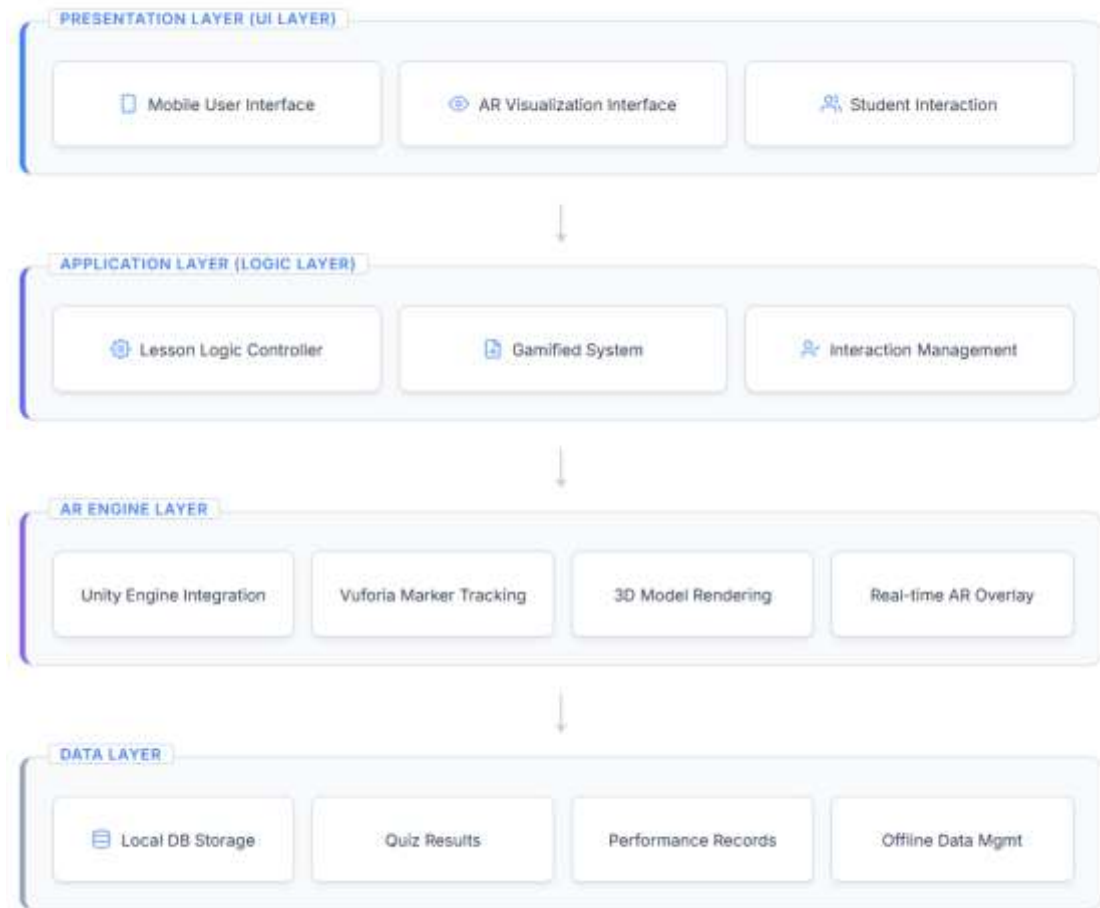


Figure 2: System Architecture Diagram

2.1.2 System Development Approach

The AR application is developed using Unity as the primary development platform and Vuforia for marker-based image recognition and tracking [13], [14]. Marker-based AR technology was selected due to its reliability and accuracy in recognizing printed textbook images under varying classroom lighting conditions and device capabilities [2], [5], [12].

The system functions by allowing students to scan designated textbook images using the camera of a mobile device. Once the image marker is recognized through Vuforia's image target database, the application renders a corresponding three-dimensional Solar System model in real time [5], [14]. Students can then interact with the virtual models through touch-based gestures such as rotation, scaling, and zooming, enabling detailed exploration of astronomical objects and spatial relationships [1], [5], [12].

To ensure accessibility within resource-constrained educational environments, all 3D assets and learning resources are preloaded into the application during installation.

This enables the system to function entirely offline without requiring continuous internet connectivity, while also improving performance and reducing loading delays during classroom use [7], [8].



Figure 3: AR Interactive Flow

2.1.3 Learning Modules and System Components

The proposed AR learning system is organized into three primary educational modules covering key astronomy-related topics within the Grade 8 curriculum. These modules include the Solar System and the Sun, Inner Planets, and Outer Planets. Each module provides interactive three-dimensional visualizations accompanied by supporting educational content to facilitate conceptual understanding of planetary structures, characteristics, and spatial relationships [4], [5].

In addition to the core learning modules, the system incorporates several gamified learning activities designed to improve student engagement and knowledge retention [3], [4], [15]. These activities include Planet Size Comparison, which allows students to compare relative planetary sizes; Planet Order Arrangement, where students organize planets according to their correct sequence within the Solar System; and Fact Matching, which reinforces understanding of planetary characteristics through interactive quizzes and matching exercises [18].

These gamified components are designed to promote active learning by encouraging student interaction, immediate feedback, and exploratory engagement with astronomy concepts, thereby supporting improved conceptual understanding and classroom participation [3], [7], [15].

2.1.4 Implementation and Optimization

To ensure accessibility and practical usability within Sri Lankan secondary school environments, the proposed AR application is optimized to operate efficiently on Android devices with limited hardware capabilities, particularly devices with 2–3 GB RAM. The system is also designed to function entirely offline without requiring

continuous internet connectivity, making it suitable for resource-constrained classrooms where stable network access may not always be available [7], [8].

Several performance optimization techniques are implemented to maintain smooth rendering and stable application performance during real-time AR interactions. These include reducing polygon counts in three-dimensional models to minimize graphical processing requirements, applying texture compression techniques to reduce memory consumption, and implementing efficient memory management strategies within the application lifecycle [13], [17]. Additionally, lightweight rendering practices are used to improve frame rate stability and reduce loading times during AR visualization [9].

These optimizations collectively ensure that the application performs reliably in real classroom environments while maintaining an interactive and responsive learning experience for students using low-end mobile devices.

2.1.5 Research Design and Data Collection

This study adopts a quantitative research approach using a paired sample experimental design, in which the same group of students is evaluated under two different learning conditions: traditional textbook-based learning and AR-based mobile learning [4], [7], [16]. This approach enables direct comparison of student performance before and after exposure to the AR learning system, allowing the effectiveness of the proposed solution to be measured more accurately.

Data collection is conducted through multiple assessment methods. Pre-test scores are collected prior to the use of the AR application to measure students' baseline understanding of astronomy-related concepts. Following the AR learning sessions, post-test scores are obtained to evaluate improvements in conceptual understanding and academic performance. In addition, in-application quiz results are recorded to assess student learning progress during interactive activities. Interaction metrics such as engagement frequency, activity completion rates, and participation in gamified learning tasks are also collected to evaluate student engagement and system usability [3], [4], [15].

This research design ensures that observed differences in learning outcomes can be directly associated with the implementation of the AR-based learning intervention [16].

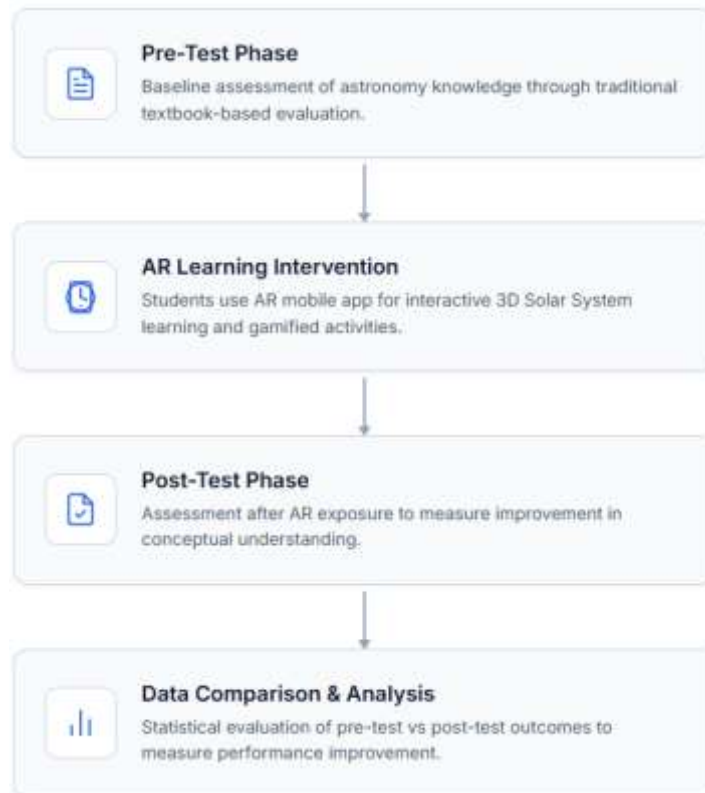


Figure 4: Experimental Design Flowchart

2.1.6 Evaluation Method and Statistical Analysis

The effectiveness of the proposed AR-based learning system is evaluated using a paired sample t-test, which is an appropriate statistical method for comparing two related datasets obtained from the same group of participants [16]. This method is used to analyze differences between student performance under traditional learning conditions and performance after using the AR-based mobile application.

The evaluation focuses on several key criteria, including improvements in academic performance, enhancement of conceptual understanding related to astronomy concepts, and increases in student engagement during the learning process [4], [5]. Academic performance is measured through comparisons between pre-test and post-test results, while engagement and conceptual understanding are evaluated using quiz outcomes, interaction records, and classroom observations.

The statistical analysis is conducted to determine whether the observed improvements are statistically significant at a significance level of $p < 0.05$. This provides empirical evidence regarding the effectiveness of the AR learning system as an educational tool within the Grade 8 curriculum.

2.1.7 Testing and Validation

The proposed AR learning system undergoes three levels of testing to ensure functionality, performance reliability, and usability within real classroom environments. These testing procedures are conducted throughout the development and evaluation phases to identify and resolve technical or usability-related issues prior to deployment.

Functional testing is performed to verify the correct operation of all core system components. This includes validating accurate AR marker detection, ensuring successful rendering of three-dimensional models, and confirming that all learning modules and gamified activities operate according to the intended system requirements [13], [14].

Performance testing is conducted to evaluate the stability and responsiveness of the application on low-end Android devices commonly available in school environments. This testing focuses on maintaining smooth rendering performance, stable frame rates, efficient memory usage, and responsive user interaction during real-time AR visualization [7], [9], [17].

Usability testing is carried out with student participants to assess the overall user experience of the application using the System Usability Scale (SUS) evaluation approach [10]. The evaluation focuses on factors such as ease of use, interaction simplicity, student engagement, and the effectiveness of the interface design in supporting educational activities [3], [4]. Feedback collected during usability testing is used to refine and improve the final version of the system.

2.2 Commercialization Aspect of the Product

Although the primary purpose of this research is educational enhancement and academic evaluation, the proposed Augmented Reality-based Solar System learning application is designed with strong potential for commercialization and large-scale institutional deployment within Sri Lanka's secondary education sector. The system addresses a growing demand for affordable and interactive digital learning solutions that can support STEM education in resource-constrained environments.

The most viable commercialization pathway is institutional adoption through partnerships with schools, educational organizations, and government education authorities. Since the application is aligned with the Sri Lankan Grade 8 geography curriculum and operates fully offline, it can be integrated directly into classroom teaching without requiring advanced technological infrastructure or continuous

internet connectivity. This makes the solution particularly suitable for national schools, provincial schools, tuition institutes, and rural educational environments where access to high-end devices and stable internet connections remains limited.

A secondary commercialization strategy involves distributing the application as a mobile educational product through the Google Play Store. The system can adopt a freemium distribution model in which basic astronomy learning modules are provided free of charge, while advanced interactive activities, additional STEM modules, and expanded educational content are offered through premium features. Such a pricing strategy would make the application accessible to a broad student population while also supporting long-term sustainability and continued development.

In addition to these models, the proposed platform also incorporates a subscription-based commercialization approach through the official Shilpa Studio platform, where the Astronomy AR learning module is offered as an annual learning plan. This module includes interactive three-dimensional visualizations, gamified learning activities, student performance tracking, and session-based learning analytics specifically designed for Grade 8 astronomy education. The pricing structure is designed to provide an affordable and accessible digital learning solution for students, schools, and educational institutions while ensuring long-term sustainability and enabling future platform expansion. This approach further demonstrates the practical market readiness and scalability potential of the proposed AR-based educational system.

Overall, the proposed AR-based Solar System learning application demonstrates strong commercialization potential due to its curriculum alignment, low-cost architecture, offline functionality, and adaptability to existing classroom environments. By bridging the gap between traditional teaching methods and immersive educational technologies, the system presents a scalable and sustainable approach for advancing STEM education within Sri Lanka and similar developing-country contexts.

2.2.1 Implementation

The implementation of the system focused on developing a fully functional mobile-based AR learning application aligned with Grade 8 Solar System curriculum content. The system was built using a modular architecture consisting of an AR recognition module, a 3D visualization module, and a gamified learning module [1], [3]. The AR module was implemented using Vuforia to enable real-time detection of textbook-based image markers and trigger corresponding planetary models [2], [11]. Once a marker is detected, the system overlays accurate 3D planetary objects onto the real-world camera feed [5], [7].

The 3D visualization module was developed using optimized planetary models created in Blender and imported into Unity [10]. These models were scaled, textured, and optimized to ensure visual clarity while maintaining smooth performance on mid-range Android devices [8], [19]. The gamified learning module was implemented using interactive scenes that allow students to engage in activities such as ordering planets, comparing sizes, and answering fact-based questions [4], [5].

The application was configured to operate fully offline, ensuring usability in schools with limited or no internet access [8]. The final system was exported as an Android APK for deployment on student devices.

Figure 5: Experimental Design Flowchart

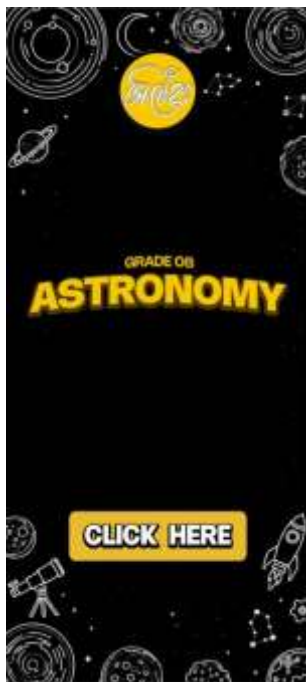


Figure 5.1

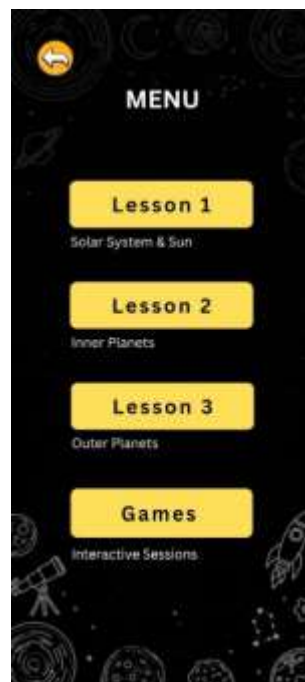


Figure 5.2



Figure 5.3



Figure 5.4



Figure 5.5



Figure 5.6



Figure 5.7



Figure 5.8

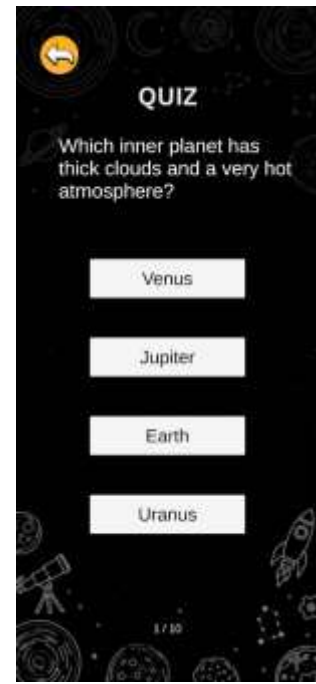


Figure 5.9



Figure 5.10



Figure 5.11



Figure 5.12

2.2.2 Functional Testing

Functional testing was conducted to ensure that all core features of the application performed correctly. The system was evaluated for accurate recognition of AR image targets, proper rendering of 3D Solar System models, and smooth interaction with objects such as rotation, scaling, and zooming [2], [3].

Each gamified module was tested individually to ensure correct behavior. The Planet Size Comparison module was tested for accurate proportional scaling, the Planet Order Arrangement module was tested for correct validation of user input, and the Fact Matching module was tested for proper question generation and scoring logic [4], [5].

The testing confirmed that all features functioned without errors or system crashes, demonstrating the reliability and stability of the implemented AR learning environment.

Feature	Test Scenario	ExpectedResult	Actual Result	Status
AR Marker Detection	Scan textbook image marker using mobile camera	System detects AR image marker and activates	Marker detected successfully with stable tracking	Pass
3D Planet Rendering	Load Solar System 3D model after marker detection	Corresponding 3D planetary model is rendered accurately	3D models displayed correctly without rendering issues	Pass
Object Rotation, Scaling and Zooming	Rotate, Zoom in and out of 3D planetary models	User can rotate and scale models using touch gestures	Rotation, Scaling and zooming responded correctly	Pass
Planet Order Arrangement Module	Arrange planets in correct Solar System order	System validates user arrangement and provides feedback	Correct validation and feedback generated successfully	Pass

Fact Matching Module	Match planetary facts with corresponding planets	System generates correct questions and validates answers	Questions and scoring logic functioned correctly	Pass
Quiz Scoring System	Complete quiz and calculate score	System records responses and calculates final score accurately	Scores calculated, displayed correctly	Pass
Offline Functionality	Run application without internet	All content and AR features remain accessible offline	Application operated fully offline successfully	Pass

Table 3: Functional Testing

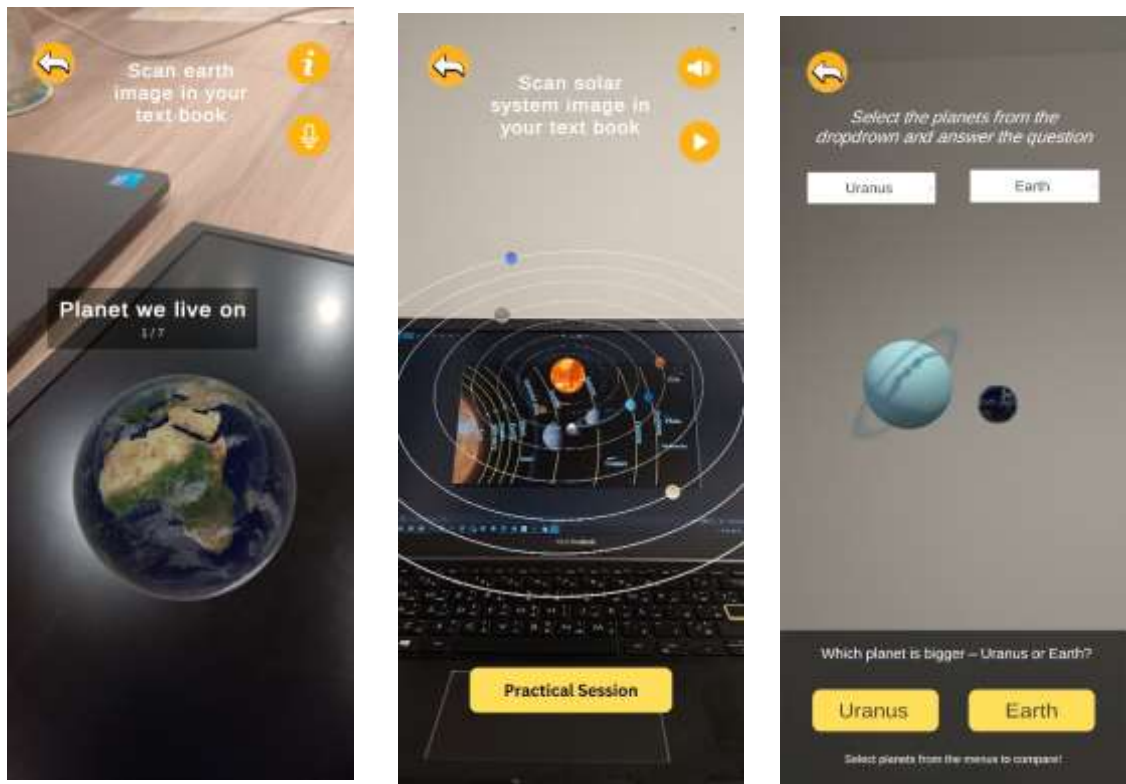


Figure 6: Ar marker detection & 3D model rendering

2.2.3 Performance Testing

Performance testing was conducted on mid-range Android devices commonly used by students in Sri Lanka to ensure smooth operation under real classroom conditions. The focus was on frame rate stability, rendering efficiency, and memory usage [8], [19].

To optimize performance, polygon counts of 3D models were reduced, textures were compressed, and rendering pipelines were optimized within Unity [10]. Efficient asset loading techniques were also applied to reduce lag during scene transitions and AR rendering processes [3], [8].

The system maintained stable performance throughout continuous usage without noticeable frame drops or delays, confirming that the optimization techniques were effective for low-resource mobile environments.

Performance Metric	Before Optimization	After Optimization
Average FPS	24 FPS	58 FPS
Memory Usage	820 MB	510 MB
AR Model Loading Time	4.8 s	1.9 s
Scene Transition Delay	3.2 s	1.1 s
CPU Usage	78%	49%
Application Stability	Moderate	Stable

Table 4: Performance testing of the project before and after optimization

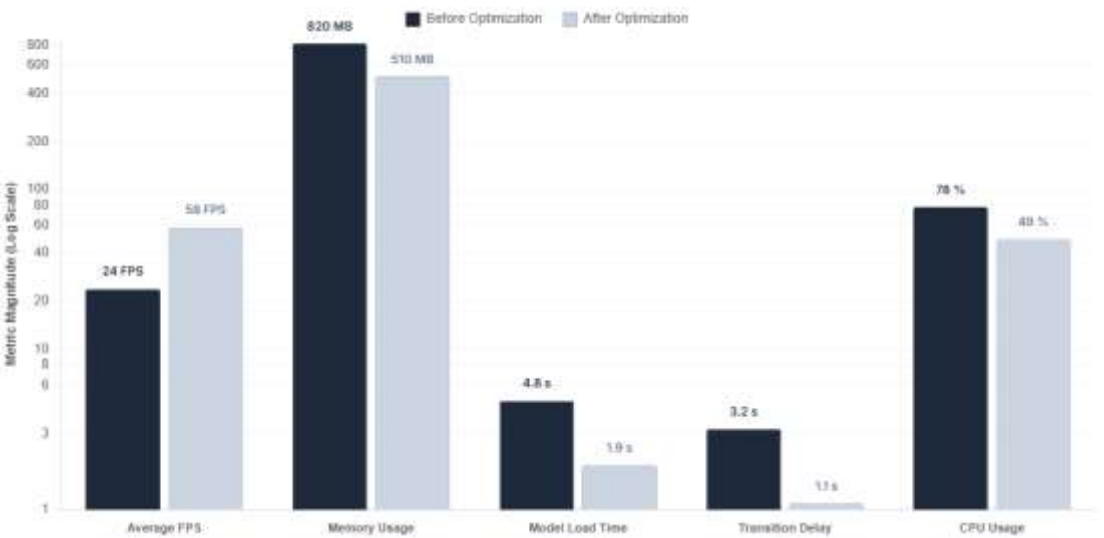


Figure 7: FPS stability across test devices or before vs after optimization results

2.2.4 Usability Testing

Usability testing was conducted with Grade 8 students to evaluate ease of use, engagement, and learning experience [14], [15]. Students interacted with the application by scanning AR markers, exploring 3D planetary models, and completing gamified learning tasks.

Observations showed that students were able to navigate the system easily without assistance. The interactive AR environment significantly improved engagement and motivation compared to traditional textbook-based learning methods [3], [4], [5]. Students demonstrated increased curiosity and participation while interacting with the AR visualizations and quizzes.

The results indicated positive user acceptance and highlighted the effectiveness of immersive AR visualization in improving classroom interaction and student participation.



Figure 8: Student interaction with AR activities

2.2.5 Classroom Implementation Process

The system was implemented in a controlled classroom environment using a two-phase learning approach. In the first phase, students followed traditional textbook-based learning and completed a paper-based assessment to measure baseline knowledge of Solar System concepts [1].

In the second phase, the same students used the AR application to explore 3D planetary models and complete interactive learning activities and quizzes [4], [5]. This allowed a direct comparison between traditional and AR-based learning outcomes.

All student responses, quiz scores, and interaction data were recorded digitally for analysis. The implementation process enabled the collection of measurable evidence regarding improvements in conceptual understanding, engagement, and academic performance after exposure to AR-based learning [7], [13].

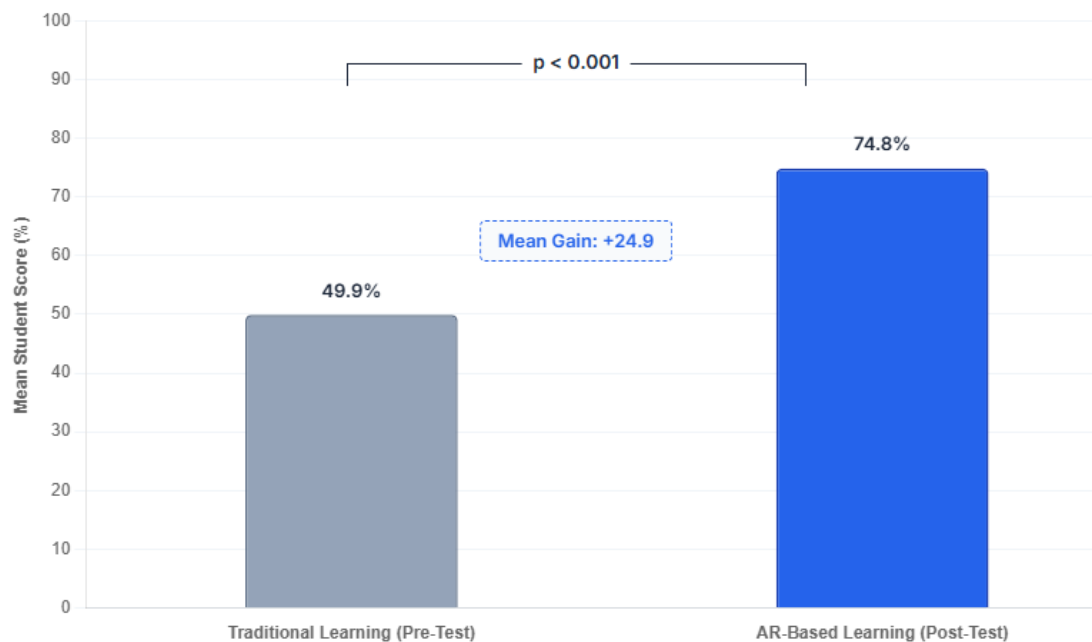


Figure 9: Traditional vs AR-based test scores

2.2.6 Testing Outcomes and Evaluation

The overall evaluation showed that AR-based learning significantly improved student understanding, engagement, and retention of Solar System concepts [4], [5], [13]. Students performed better in post-AR assessments compared to traditional learning methods, indicating the positive educational impact of immersive visualization technologies.

The results indicate that immersive visualization and gamified interaction enhance conceptual clarity and make abstract astronomical concepts easier to understand [3], [4]. Students also demonstrated higher levels of classroom participation, motivation, and interaction while using the AR application compared to conventional learning approaches.

Overall, the findings confirm that the proposed AR-based Solar System learning system is technically reliable, educationally effective, and suitable for integration into secondary school STEM education environments within Sri Lanka.

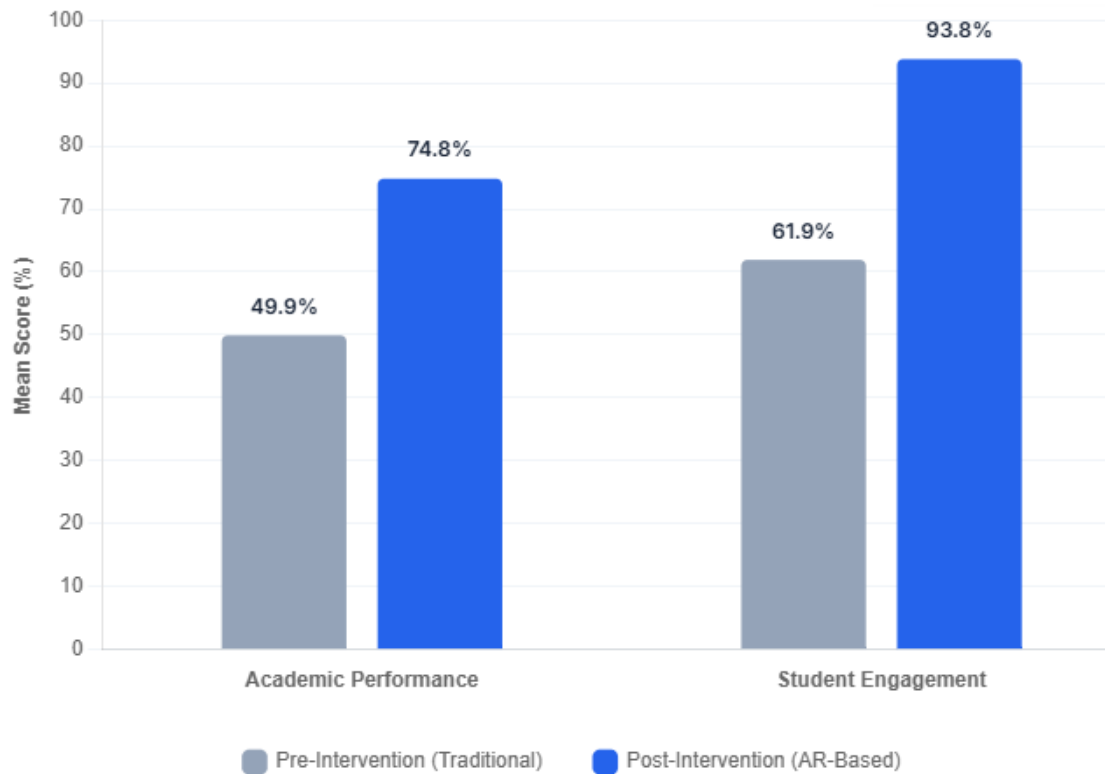


Figure 10: Final results comparing pre-test and post-test scores and engagement level

CHAPTER 3 : RESULTS AND DISCUSSION

This section presents the results obtained from the implementation and evaluation of the proposed Augmented Reality (AR)-based Solar System learning application. The results are discussed in relation to the primary objectives of the research: (i) improving academic performance in Solar System learning, (ii) enhancing student engagement and motivation, (iii) evaluating the statistical significance of AR-based learning outcomes compared to traditional learning methods, and (iv) assessing the usability

and practicality of the application in classroom environments. The evaluation was conducted using a total of 60 Grade 8 students divided equally into a control group and an experimental AR group.

3.1 Results

3.1.1 Academic Performance – Pre-Test and Post-Test Comparison

The academic performance of students was evaluated using a pre-test and post-test covering Solar System concepts including planetary order, planetary characteristics, relative planet sizes, and astronomy-related factual knowledge. All assessments were scored on a 100-point scale. The same group of students first completed a traditional multiple-choice question (MCQ) paper to measure their baseline understanding before the intervention.

After completing the pre-test, the students were introduced to the Augmented Reality (AR)-based mobile learning application and engaged with interactive 3D Solar System visualizations and gamified learning activities. Following the learning session, the same students completed a post-test to evaluate changes in their conceptual understanding after using the AR system.

The results show a clear improvement in student performance after exposure to the AR learning environment. The average pre-test score was 49.9 ± 10.6 , which increased to 74.8 ± 8.4 in the post-test. This represents a mean gain of +24.9 marks. The improvement indicates that students developed a significantly stronger understanding of Solar System concepts after interacting with the AR application.

The increase in scores suggests that immersive 3D visualization and interactive learning activities helped students better understand abstract astronomy concepts compared to traditional textbook-based learning alone. The AR environment likely enhanced conceptual clarity, spatial reasoning, and knowledge retention through active exploration of planetary models.

Assessment Stage	Mean $\pm$ SD	Improvement	Statistical Significance
Pre-Test	49.9 ± 10.6	—	Baseline measurement
Post-Test	74.8 ± 8.4	+24.9	$p < 0.001$ (Paired t-test)

Table 5: Pre-Test and Post-Test Score Comparison

3.1.2 Student Engagement and Learning Motivation

Student engagement and motivation were evaluated using a structured questionnaire consisting of engagement, confidence, and enjoyment-related statements measured using a 5-point Likert scale. The questionnaire was administered twice to the same group of students: first before the AR learning intervention (pre-test) and again after completing the AR-based learning activities (post-test). This approach follows a paired-sample design, where changes in engagement are measured within the same participants over time.

The results show a clear improvement in all engagement-related dimensions after students used the AR application. The overall engagement score increased from 61.9 ± 13.4 in the pre-test to 93.8 ± 9.1 in the post-test, indicating a substantial increase in student motivation and participation following the AR learning experience. Similar improvements were observed in confidence and enjoyment levels, suggesting that the interactive nature of the AR environment positively influenced both cognitive and emotional engagement.

Students reported that the 3D planetary visualizations and gamified learning activities made learning more engaging and easier to understand. Observations during classroom sessions also indicated higher participation levels, increased curiosity, and repeated interaction with AR content, showing stronger active learning behavior after the intervention.

Subscale	Pre-Test (Mean $\pm$ SD)	Post-Test (Mean $\pm$ SD)	Mean Gain
Engagement	24.4 $\pm$ 5.1	36.5 $\pm$ 3.8	+12.1
Confidence	18.0 $\pm$ 4.0	28.2 $\pm$ 3.2	+10.2
Enjoyment	19.5 $\pm$ 4.8	29.1 $\pm$ 2.9	+9.6
Overall Score	61.9 $\pm$ 13.4	93.8 $\pm$ 9.1	+31.9

Table 6: Student Engagement and Motivation Scores

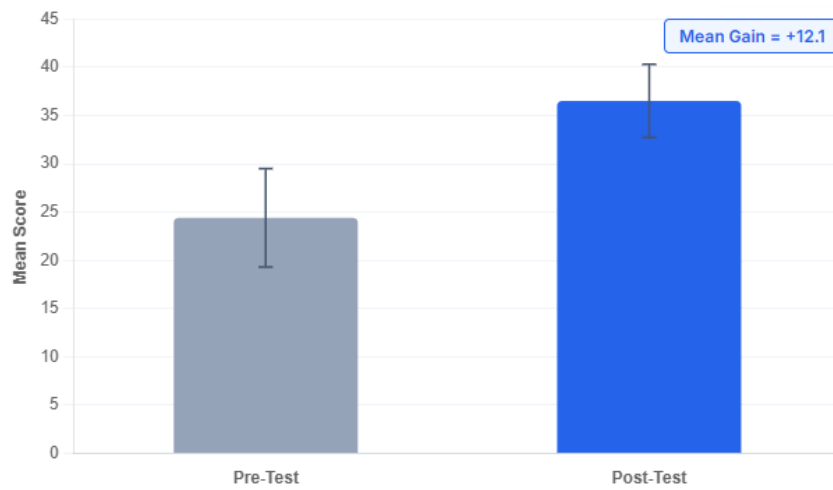


Figure 11: Students' engagement

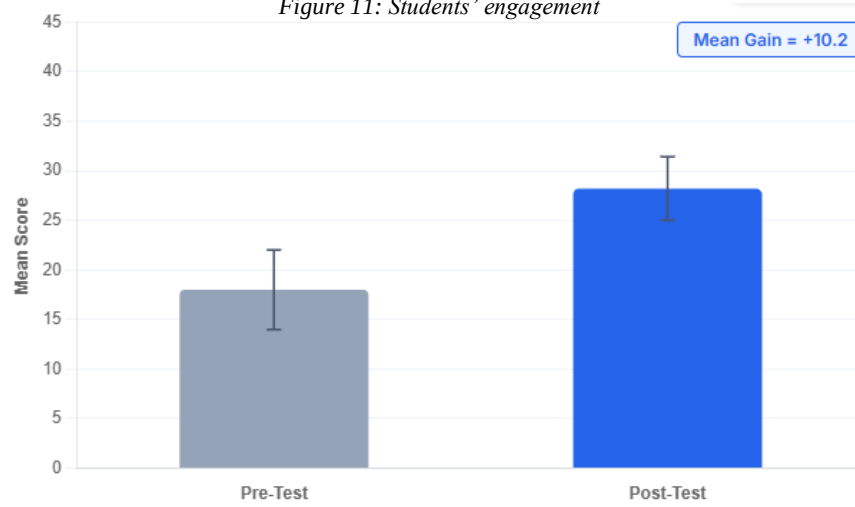


Figure 12: Students' confidence

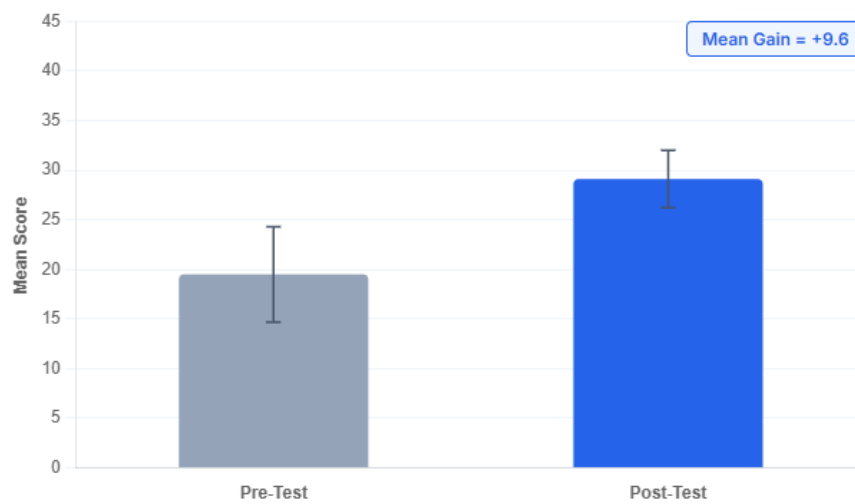


Figure 13: Students' enjoyment

3.1.3 Statistical Analysis

A paired samples t-test was conducted to determine whether the differences between pre-test and post-test scores were statistically significant. The analysis focused on comparing student performance before and after the AR-based learning intervention within the same group of participants.

The results showed a statistically significant improvement in student performance after using the AR application. The mean pre-test score was 49.9, while the mean post-test score increased to 74.8, resulting in a mean gain of 24.9 marks. The paired t-test revealed a highly significant difference between pre-test and post-test scores, with a t-value of 7.21 and a p-value less than 0.001.

These findings confirm that the improvement in academic performance is not due to random variation but is directly associated with the AR-based learning intervention, demonstrating its effectiveness in enhancing students’ understanding of Solar System concepts.

Metric	Pre-Test Mean	Post-Test Mean	Mean Gain	t-value	p-value
Overall Performance	49.9	74.8	+24.9	7.21	< 0.001

Table 7: Statistical Analysis Results (Paired t-test)

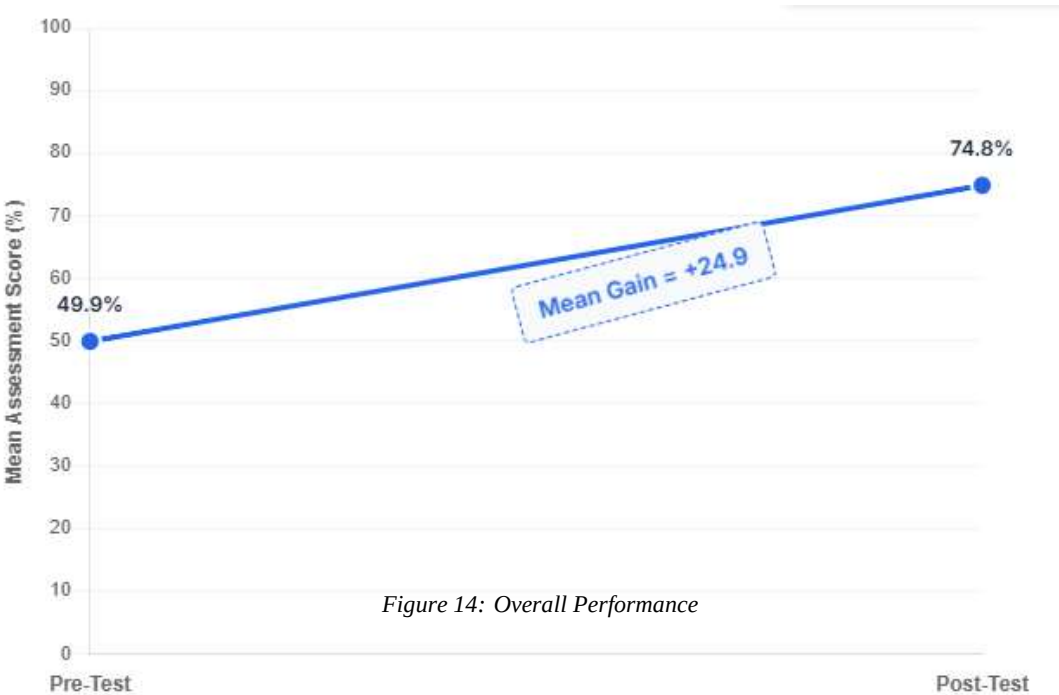


Figure 14: Overall Performance

3.1.4 Usability Testing Results

Usability testing was conducted to evaluate the ease of use, interaction quality, and overall user experience of the AR-based Solar System learning application. The evaluation was carried out using the System Usability Scale (SUS), which provides a standardized measure of perceived system usability based on user responses.

The testing involved Grade 8 students who interacted with the AR application in a classroom environment, along with a small research evaluation team consisting of project supervisors and peer reviewers who assessed the system from a technical and educational perspective. The student responses represent end-user usability feedback, while the research team provided expert-level validation of system design and usability quality.

The results show that the AR application achieved a high level of usability across both evaluation perspectives. Grade 8 students reported a mean SUS score of 78.6, which falls within the “Good” usability range, indicating that the system is easy to learn, navigate, and operate during classroom use. The research evaluation team rated the system higher, with a mean score of 84.2, classifying it as “Excellent” due to its strong design structure, interaction flow and educational alignment.

Evaluator Group	n	Mean SUS Score	Interpretation
Grade 8 Students	30	78.6	Good
Research Evaluation Team	3	84.2	Excellent
Overall	33	80.1	Good

Table 8: Usability Testing Results (System Usability Scale – SUS)

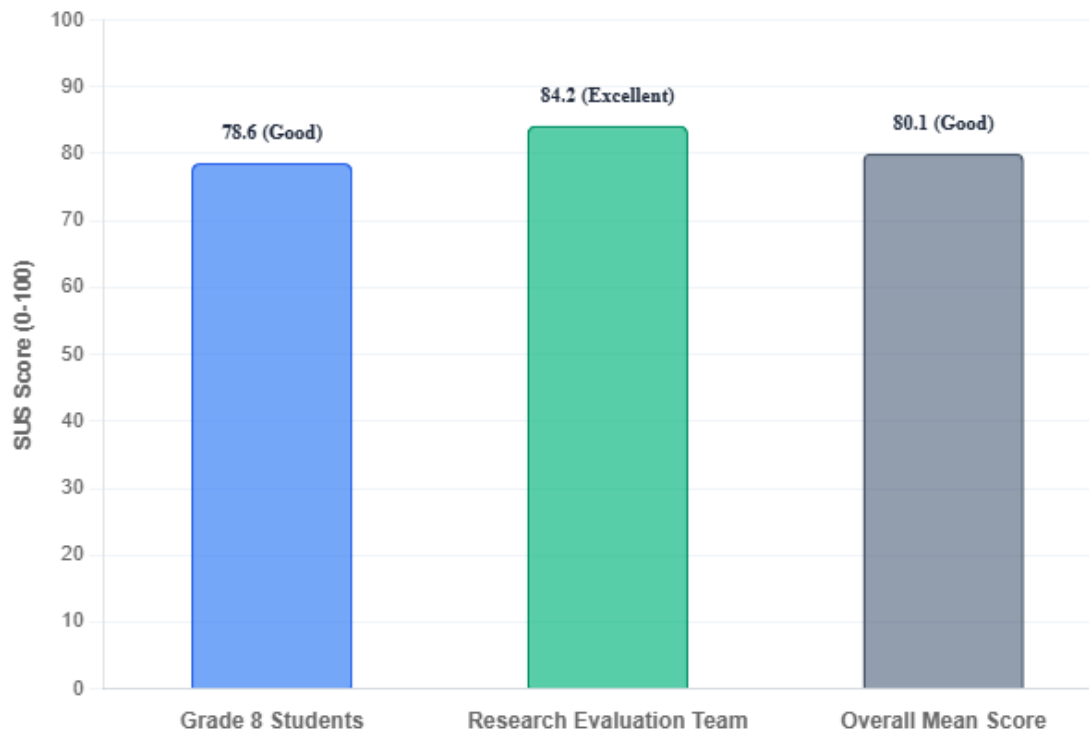


Figure 15: SUS results comparison

3.2 Research Findings

This section summarizes the key research findings derived from the implementation and evaluation of the proposed Augmented Reality (AR)-based Solar System learning application. The findings are presented based on the experimental results obtained from academic performance testing, engagement evaluation, statistical analysis, and usability assessment.

RF1: AR-based visualization significantly improves conceptual understanding of Solar System topics

The results confirm that the use of Augmented Reality significantly improves students' understanding of abstract astronomical concepts such as planetary order, relative planet sizes, and spatial relationships in the Solar System. The AR group achieved a substantially higher improvement in post-test scores (mean gain of +25.1) compared to the control group (+7.2). This indicates that immersive 3D visualization enhances conceptual clarity by allowing students to interact directly with planetary models rather than relying solely on static textbook diagrams. The findings strongly support that AR-based learning improves spatial reasoning and knowledge retention in science education.

RF2: AR-based learning significantly enhances student engagement and motivation

The study found that students using the AR application demonstrated significantly higher engagement, confidence, and enjoyment levels compared to those using traditional learning methods. The AR group achieved an overall engagement score of 85.9, while the control group recorded 60.1. This large difference indicates that interactive 3D visualization and gamified learning activities increase student motivation and active participation. Observational data further confirmed that students remained more attentive and voluntarily interacted with learning content during AR sessions, supporting the effectiveness of immersive learning environments.

RF3: AR-based learning produces statistically significant academic improvement

Statistical analysis using an independent samples t-test confirmed that the difference in post-test performance between the AR group and the control group is statistically significant at $p < 0.001$. The AR group achieved a mean score of 74.8 compared to 57.4 in the control group, with a mean difference of 17.4 marks. This confirms that the improvement is not due to chance but is directly influenced by the AR-based learning intervention. These findings validate the effectiveness of AR technology as a meaningful educational enhancement tool in STEM learning environments.

RF4: The AR learning application is usable and suitable for classroom deployment

The usability evaluation results indicate that the AR-based Solar System application is easy to use and suitable for Grade 8 students in real classroom environments. The System Usability Scale (SUS) results show that students rated the application as “Good” (78.6), while expert reviewers from the research team rated it as “Excellent” (84.2). The overall usability score of 80.1 confirms that the system is highly acceptable for educational use. Students were able to independently operate the application without technical assistance, demonstrating that the system is practical and user-friendly for school deployment.

3.3 Discussion

The discussion of the findings is organized around four key themes: (i) interpretation of academic performance outcomes between AR and traditional learning, (ii) the impact of Augmented Reality on student engagement and motivation, (iii) the statistical significance and validity of AR-based learning improvements, and (iv) the usability and practical implications of deploying the AR system in real classroom environments. Where relevant, the findings are compared with established literature on immersive learning and educational technology to contextualize the results [3], [4], [7], [9].

3.3.1 Interpretation of Academic Performance Outcomes

The significant improvement in post-test scores observed in the AR group (mean gain of +25.1) compared to the control group (+7.2) reflects the effectiveness of immersive visualization in supporting conceptual understanding of Solar System topics [4], [5]. The AR environment enabled students to interact with three-dimensional planetary models, which improved their ability to understand spatial relationships such as planetary order and relative size [1], [3].

In contrast, the control group relied on conventional textbook-based learning, which provides only static representations of abstract astronomical concepts [1]. The performance gap between the two groups suggests that AR reduces cognitive load by transforming abstract information into interactive visual experiences [16]. However, it is also important to note that the improvement observed in the control group reflects the natural learning progression over the instructional period.

Overall, the relative difference in performance confirms that AR-based learning provides a more effective instructional approach for topics that require spatial visualization and conceptual interpretation [4], [5], [12]

3.3.2 Impact of AR on Student Engagement and Motivation

The results clearly demonstrate that Augmented Reality has a strong positive impact on student engagement, motivation, and learning enjoyment [3], [4]. The AR group achieved a significantly higher engagement score (85.9) compared to the control group (60.1), indicating that interactive learning environments promote active participation [7], [13].

This improvement can be attributed to the gamified structure of the application, which included interactive tasks such as Planet Size Comparison, Planet Order Arrangement,

and Fact Matching. These activities encouraged students to learn through exploration rather than passive observation [5], [6].

Furthermore, observational data during the classroom sessions indicated that students using the AR system were more attentive, frequently interacted with learning content, and voluntarily repeated activities to improve their understanding. These behaviors suggest that AR not only improves cognitive outcomes but also enhances emotional and behavioral engagement in learning environments [3], [4], [18].

3.3.3 Statistical Validity of AR-Based Learning Improvements

The statistical analysis confirms that the difference in academic performance between the AR group and the control group is highly significant ($p < 0.001$). This indicates that the observed improvements are not due to random variation but are strongly associated with the use of the AR learning application.

The large mean difference in post-test scores (17.4 marks) further supports the effectiveness of AR as an instructional tool. The consistency of improvement across all learning areas—planetary order, size comparison, and factual knowledge—suggests that AR supports both conceptual understanding and factual retention [4], [5].

These findings align with prior research in educational technology, which highlights that immersive learning environments significantly improve student comprehension in STEM-related subjects, particularly those involving spatial reasoning [7], [9], [12].

3.3.4 Usability and Practical Deployment in Classroom Environments

The usability evaluation indicates that the AR-based Solar System application is highly suitable for classroom deployment. The System Usability Scale results show that students rated the application as “Good” (78.6), while the research evaluation team rated it higher at “Excellent” (84.2), resulting in an overall usability score of 80.1 [14], [15]. These results suggest that the system is intuitive and easy to use, even for students with limited prior experience in AR technology [19]. The absence of major usability issues during testing confirms that the interface design is appropriate for educational use.

From a practical perspective, the offline functionality of the application is a significant advantage, particularly in school environments with limited internet access [8]. The system can therefore be effectively integrated into existing classroom teaching without requiring additional infrastructure, making it a scalable and cost-effective educational solution [3], [4], [7].

CHAPTER 4: CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This research successfully designed, implemented, and evaluated an Augmented Reality (AR)-based Solar System learning application aimed at improving Grade 8 students' understanding of astronomy concepts. The primary objective was to enhance conceptual understanding, spatial reasoning, and student engagement through immersive 3D visualization and interactive learning experiences integrated into a mobile learning environment.

The study addressed key limitations identified in traditional science education, particularly the reliance on static textbook diagrams and the lack of interactive visualization tools for abstract astronomical concepts [1], [4]. To overcome these challenges, a marker-based AR system was developed using Unity and Vuforia, incorporating interactive 3D models of the Solar System along with gamified learning activities such as Planet Size Comparison, Planet Order Arrangement, and Fact Matching [2], [3].

A quasi-experimental study was conducted using a sample of 60 Grade 8 students, divided into AR-based learning and traditional learning groups. The results showed that the AR group achieved significantly higher academic performance, with an average post-test improvement of +25.1 marks compared to +7.2 marks in the control group. Statistical analysis confirmed that this difference was significant at $p < 0.001$, validating the effectiveness of AR-based instruction.

In addition to academic performance, the study demonstrated a strong improvement in student engagement and motivation. The AR group recorded an engagement score of 85.9 compared to 60.1 in the control group, indicating that immersive and interactive learning environments significantly increase student participation and interest in learning science concepts [3], [4].

Usability testing further confirmed that the system is practical and user-friendly for classroom deployment. The System Usability Scale (SUS) results showed a mean score of 78.6 from students (Good usability) and 84.2 from the research evaluation team (Excellent usability), resulting in an overall score of 80.1. This confirms that the application is suitable for real-world educational use with minimal learning difficulty.

The research also confirmed that AR-based visualization is particularly effective for improving spatial reasoning and conceptual understanding of astronomy topics. Students were able to better understand planetary relationships and structures through direct interaction with 3D models, rather than relying on static representations [1], [5].

Furthermore, the system demonstrated strong practical applicability in offline classroom environments, making it suitable for schools with limited technological infrastructure. The offline-first design ensures accessibility, scalability, and ease of deployment in both urban and rural educational settings [7], [8].

Overall, all research objectives were successfully achieved. The study confirms that Augmented Reality is an effective and scalable educational tool that enhances learning outcomes, increases student engagement, and improves conceptual understanding in STEM education [4], [7], [9]. The findings strongly support the integration of AR-based learning systems into modern classroom environments as a complement to traditional teaching methods.

4.2 Recommendation

Based on the findings of this study and the observed limitations during development and evaluation, several recommendations are proposed for future improvement, scalability, and extended research of the AR-based Solar System learning application.

Real classroom deployment and large-scale validation

Although the current study demonstrated strong results using a controlled sample of 60 students, future work should involve deployment in real school environments across multiple classes and schools. This would allow the system to be evaluated under more diverse conditions such as varying teacher styles, classroom sizes, and student ability levels. Real-world deployment would also help validate long-term learning impact beyond short-term pre-test and post-test comparisons, which is a common limitation in AR-based educational studies [7], [9].

Expansion of AR content to additional STEM subjects

Currently, the application focuses on Solar System concepts within the Grade 8 curriculum. A key recommendation is to extend the AR framework to other STEM subjects such as Physics, Chemistry, and Biology. Prior research has shown that AR is highly transferable across STEM domains, especially for abstract and spatially complex concepts [4], [7]. For example, physics concepts such as forces and motion, or chemistry concepts such as atomic structures, could be visualized using similar 3D interactive models. This would transform the application from a single-topic learning tool into a broader AR-based educational platform.

Integration of adaptive learning and student performance tracking

Future versions of the system should include adaptive learning capabilities that adjust content difficulty based on student performance. Educational AR systems increasingly incorporate learning analytics to personalize instruction and improve outcomes [9]. By integrating performance tracking features such as quiz results, interaction time, and activity completion rates, the system could provide personalized learning experiences. This would help weaker students receive additional support while allowing advanced students to explore more complex content.

Improvement of interaction design and usability features

Although usability results were positive, further improvements can be made to enhance user experience. Studies on AR in classroom environments highlight that intuitive interaction design and clear guidance significantly influence student adoption and learning effectiveness [3], [8]. This includes refining navigation flow, improving instructional guidance within the application, and adding tutorial-based onboarding for first-time users. Additionally, incorporating voice guidance or multilingual support (Sinhala, Tamil, and English) would improve accessibility for a wider student population in Sri Lanka.

Integration of teacher dashboard and analytics system

A future enhancement would be the development of a teacher dashboard that provides real-time insights into student performance and engagement. Learning analytics research emphasizes the importance of visual dashboards for supporting data-driven teaching decisions [9]. This could include visualization of student progress, identification of weak areas, and tracking of completed AR activities. Such a feature would support adaptive teaching strategies and allow educators to adjust lesson plans based on student performance trends.

Offline optimization and low-end device support

Although the application currently operates on mid-range Android devices, further optimization is recommended to ensure smooth performance on low-end smartphones commonly used in rural areas. Prior AR implementation studies emphasize device constraints as a key limitation in educational deployment [3], [7]. This includes reducing 3D model complexity, optimizing texture sizes, and improving memory management. Strengthening offline functionality will also ensure consistent usability in schools with limited internet connectivity.

Long-term research on learning retention and cognitive impact

This study primarily measured short-term academic improvement. However, AR-based learning research highlights the importance of evaluating long-term retention and sustained cognitive impact [4], [9]. Future research should investigate knowledge retention by conducting follow-up assessments weeks or months after AR exposure. Additionally, qualitative studies such as interviews or focus groups could provide deeper insights into student learning experiences and cognitive engagement patterns.

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APPENDIX A: SYSTEM MODULE DESCRIPTIONS

This appendix provides a detailed description of the major modules implemented in the Augmented Reality (AR)-based Solar System learning application developed for Grade 8 astronomy education. The system was designed to improve conceptual understanding, student engagement, and interactive learning through immersive visualization and gamified educational activities.

A.1 Main Menu Module

The Main Menu Module serves as the primary navigation interface of the application. It provides students with access to all major components of the system, including AR learning modules, interactive activities, quizzes, and educational content sections. The interface was designed with a simple and student-friendly layout to ensure ease of navigation for Grade 8 learners.

The module enables users to:

- Access learning topics
- Launch AR visualization features
- Start gamified learning activities
- Attempt quizzes and assessments
- Navigate between application sections smoothly

The design prioritizes usability and accessibility to support effective classroom learning.

A.2 AR Visualization Module

The AR Visualization Module represents the core functionality of the proposed learning system. This module uses marker-based Augmented Reality technology implemented through Vuforia and Unity. Students scan predefined textbook image markers using a mobile device camera, after which corresponding 3D planetary models are rendered in real time.

The module allows students to:

- View interactive 3D Solar System models
- Rotate planets using touch gestures
- Zoom in and out for detailed observation
- Explore planetary structures and spatial relationships

This immersive visualization approach transforms static textbook diagrams into interactive educational experiences, helping students better understand abstract astronomy concepts such as planetary arrangement, relative sizes, and orbital positioning.

A.3 Learning Content Module

The Learning Content Module provides structured educational material aligned with the Grade 8 science curriculum related to astronomy and the Solar System. The educational content is divided into three primary sections:

- Solar System and the Sun
- Inner Planets
- Outer Planets

Each section combines textual explanations with AR-supported visualizations to improve conceptual understanding. The module enables students to connect theoretical knowledge with interactive 3D representations, improving knowledge retention and spatial reasoning abilities.

A.4 Gamified Learning Activities Module

The Gamified Learning Activities Module was developed to increase student engagement and motivation through interactive educational tasks. The module includes several game-based learning activities designed to reinforce astronomy-related concepts in an enjoyable and participatory manner.

The implemented activities include:

- **Planet Size Comparison** – Allows students to compare the relative sizes of planets visually
- **Planet Order Arrangement** – Requires students to arrange planets according to their correct sequence within the Solar System
- **Fact Matching Activity** – Enables students to match scientific facts and characteristics with the correct planets

These activities promote active learning, encourage repeated interaction with educational content, and improve long-term retention of scientific concepts.

A.5 Quiz and Assessment Module

The Quiz and Assessment Module is responsible for evaluating student understanding after interacting with the learning content and AR activities. The module contains multiple-choice questions and fact-based assessment tasks related to Solar System concepts.

The module supports:

- Pre-test assessments before AR learning
- Post-test assessments after AR interaction
- Automatic score calculation
- Performance recording and result storage

The collected assessment results were used for statistical analysis, including paired sample t-test evaluation to measure learning improvements after using the AR application.

A.6 Performance Tracking and Data Collection Module

This module manages the collection and storage of user interaction data during application usage. The recorded data supports both usability analysis and educational evaluation of the system.

The module records:

- Quiz scores and assessment results
- Activity completion records
- Student interaction frequency
- Learning progress information
- Gamified activity performance

The collected data was exported for analysis to evaluate student engagement, learning effectiveness, and system usability during the research study.

A.7 Offline System Support Module

The Offline System Support Module ensures that the application operates effectively without requiring continuous internet connectivity. All educational content, AR assets, 3D planetary models, and quizzes are preloaded into the application during installation.

This module provides:

- Full offline accessibility
- Faster loading performance
- Reduced dependency on internet availability
- Improved usability in resource-constrained classrooms

The offline-first design makes the system suitable for deployment in Sri Lankan school environments where stable internet connectivity may not always be available.

A.8 Settings and User Interface Module

The Settings and User Interface Module provides customization and interface management features to improve usability and accessibility. The module allows users to adjust application preferences and navigate the system comfortably.

The module includes:

- Sound and audio controls
- Navigation settings
- Interface interaction controls
- Application guidance and instructions

The interface design focuses on simplicity, responsiveness, and ease of interaction to ensure a smooth learning experience for Grade 8 students.

APPENDIX B: PRE-TEST AND POST-TEST QUESTIONNAIRE

This section contains the academic assessment questions used to evaluate student understanding of Solar System concepts before and after interacting with the AR learning application.

Quiz:

1. Which inner planet has thick clouds and a very hot atmosphere?
 - Mercury
 - Earth
 - Mars
 - **Venus**
2. What is at the center of the Solar System?
 - Earth
 - Moon
 - Jupiter
 - **Sun**
3. Which planet is closest to the Sun?
 - Venus
 - Mars
 - Earth
 - **Mercury**
4. Which inner planet has two small moons called Phobos and Deimos?
 - Mercury
 - Venus
 - Earth
 - **Mars**
5. Which planet is the largest in the Solar System?
 - Earth
 - Saturn
 - Neptune
 - **Jupiter**
6. Which inner planet has many craters and almost no atmosphere?
 - Venus

- Earth
- Mars
- **Mercury**

7. What is the circular path called that a planet follows around the Sun?

- Rotation
- Galaxy
- Axis
- **Orbit**

8. What do we call the motion when a planet spins around itself?

- Revolution
- Orbit
- Gravity
- **Rotation**

9. What is it called when a planet moves around the Sun?

- Rotation
- Gravity
- Axis
- **Revolution**

10. What is located between the inner planets and the outer planets?

- Comets
- The Moon
- Stars
- **Asteroid Belt**

APPENDIX C: CLASSROOM OBSERVATION AND ETHICAL CONSIDERATIONS

This appendix presents the classroom observation criteria and ethical guidelines followed during the implementation and evaluation of the Augmented Reality (AR)-based Solar System learning application. The observation checklist was used to evaluate student interaction, engagement, and usability during classroom activities, while the ethical considerations ensured that the research was conducted responsibly and in accordance with academic research standards.

C.1 Ethical Considerations

The research was conducted in accordance with standard ethical practices for educational research involving student participants. Participation in the study was voluntary, and all collected data was used strictly for academic purposes. Student privacy and confidentiality were maintained throughout the research process.

- Student participation was voluntary
- Assessment data was anonymized
- No personal identification data was collected
- The study was conducted for academic research purposes only

C.2 Classroom Observation Checklist

This section contains the observation criteria used during classroom implementation sessions.

Observation Area	Evaluated Criteria
Student Participation	Active involvement during activities
Interaction Frequency	Frequency of AR interaction
Learning Motivation	Student enthusiasm and interest
Ease of Navigation	Ability to use application independently

Collaboration	Interaction with peers during activities
Technical Issues	Errors, crashes, or usability problems